

CONFIGURATION AND SECURITY DOCUMENTATION

Project Performance Improvement Toolkit

Variables, secrets, access, and data protection

Technical source: oli_v6_deploy.py

Updated: May 11, 2026

1. Required Configuration

Element	Expected Value
Local command	streamlit run oli_v6_deploy.py
AI variable	OPENAI_API_KEY available as an environment variable.
Production mode	STREAMLIT_ENV=production when paths are read from st.secrets.
Execution directory	Directory containing the required Excel, CSV, and .pt files unless production paths are used.
Dependencies	Installed from requirements.txt.

2. Secrets and Paths

The code reads OPENAI_API_KEY with os.getenv. In Streamlit Cloud or another host, the key must be exposed as an effective environment variable for the Python process.

Name	When Used	Expected Content
df_path	Production	Accessible path to df_complete_all_full.xlsx.
df_raw_path	Production	Accessible path to df_split_actions.xlsx.
embeddings_path	Production	Accessible path to emb_Recomm_rec_cl_4.pt.
structured_embeddings_path	Production	Accessible path to Recommendation_RAG_Metadata.pt.
analyzed_recommendations_path	Optional production	Accessible path to analyzed_recommendations_plans_v5.csv.
lessons_embeddings_path	Production for inactive helper	Accessible path to emb_LL_IL_cl_4.pt if the lessons function is reconnected.
structured_lessons_path	Production for inactive helper	Accessible path to lessons_metadata.pt if the lessons function is reconnected.

3. Access Control

oli_v6_deploy.py does not implement users, roles, passwords, feature-level permissions, audit, tenant isolation, uploaded-file scanning, or secret redaction. Controls must be applied outside the app through Streamlit Cloud, an institutional server, proxy, private network, or equivalent mechanism.

- Only authorized users should be able to open the application.
- The repository and data files should have separate permissions: read for operation, write only for maintenance.
- Access to OPENAI_API_KEY should be limited to the runtime environment and technical administrators.
- Generated downloads are the responsibility of the user who creates them.

4. Data Security

- Uploaded documents are processed in memory and through temporary files; temporary files are deleted after extraction when the flow reaches cleanup.
- Document content, questions, and evidence can be sent to OpenAI for evaluations and answers.
- st.session_state retains extracted text and results during the active session.
- Downloadable XLSX/ZIP files can contain verbatim quotes, reasoning, and evidence from the source document.
- Local pickle caches, including embeddings_cache.pkl and analysis_cache.pkl, are neither encrypted nor integrity-checked.
- The application does not encrypt local outputs or implement retention policy; those policies must be defined in the institutional environment.

4.1 Active Cache Layers

Layer	Location	Operational Risk
st.cache_data	Streamlit memory/process	Can retain previous files, rubrics, or data until cache is cleared or the process restarts.
Tab 4	st.session_state	Stores documents, chunks, and embeddings by session; content changes with the same file name may not be detected.
Tabs 5 and 6	embeddings_cache.pkl	Persistent local cache, without encryption or integrity control.
Deep analysis	analysis_cache.pkl	Persistent local cache, without encryption or integrity control.
SimpleHierarchicalStore	~/document_store/ embedding_cache.pkl	Legacy helper for hierarchical evaluation; not the main Tab 4 mechanism.

5. Safe Operation Parameters

Parameter	Value in Code	Impact
Global MAX_WORKERS for preliminary appraisal	48	High question concurrency; requires sufficient API limits.
MAX_WORKERS Tab 2	3	Reduces rate-limit risk in rubric

		evaluation.
MAX_WORKERS Tab 3	8	Processes sustainability criteria in parallel.
Chat RAG	>100,000 characters	Avoids sending very large documents in full for every query.
Token limit	110,000 tokens	Cuts very long documents before model calls.
Classification batch	20 texts per logical batch	Controls embedding generation for unique recommendations.

6. Pre-Production Checklist

- Confirm OPENAI_API_KEY exists in the runtime process and the account has sufficient budget.
- Confirm all mandatory data files open with pandas/torch in production.
- Configure private access to the app in the host.
- Run one test per tab with non-sensitive documents.
- Verify downloads are generated, open correctly, and do not expose more information than needed.
- Internally document who may replace data files and who may deploy code.

7. Incident Response

1. Disable external access to the app from the host or proxy.
2. Revoke and rotate OPENAI_API_KEY if exposure is suspected.
3. Preserve copies of downloaded files or available logs only if institutional policy permits.
4. Restore data files from a verified backup.
5. Reactivate the app after a complete Tabs 1-6 test.